

**North Dakota State University
Department of Computer Science**

CSci 313: Software Development with Frameworks

Spring 2026

3 Credits

Section – 3 (Online, Asynchronous)

Instructor: **Pratap Kotala, Ph.D.**

Office Address: QBB 258-A26,

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OFFICE (STUDENT) HOURS

- The student hours would be on the following times and days in QBB 258-A26 and over Zoom:
 - 1:00 pm – 2:00 pm: Monday, and Wednesday
 - 12:00 pm – 1:00 pm: Thursday
 - Or appointment via email
- A link to the Zoom Office Hours is available on Blackboard.
- If you are busy during office hours, please email the instructor so a Zoom meeting can be set up at a mutually convenient time.

Learning Assistants:

Tutoring Hours – TBD. Will be communicated through Blackboard announcement.

BULLETIN DESCRIPTION:

This course provides students with an understanding of the full-stack development of real-world software applications using software frameworks. Students will work in small teams to use an agile methodology to develop a full-stack application.

Pre-Requisites – CSci 161

COURSE OBJECTIVES:

Modern software development makes wide use of software frameworks to create high-quality software applications. This course will cover the full-stack development of real-world software applications using software frameworks. The students will gain an understanding and insight into the practical use of modern and relevant software architectures and frameworks through group projects on the various frameworks using an agile methodology. Students will learn how

requirements specifications and business analysis combined with modern frameworks play an important role in the successful completion of a software project.

Learning Objectives –

Upon successful completion of the course, the student will be able to:

- Gain insights into the practical use of software frameworks and architectures in modern software development
- Understand the full-stack development including front and back-end development process.
- Design and develop modern, complex, responsive, and scalable web applications.
- Follow agile software development practice in the practical software development process.
- Learn to work in teams cooperatively on their projects helping students to hone communication skills including the ability to discuss and compromise where there are differences of opinion on how to proceed with the task.

COURSE DELIVERY

This course is presented in an asynchronous, online format through Blackboard. There will be no mandatory class meetings. The course materials, including PowerPoint slides, recorded video lectures, and additional resources, can be accessed on the Blackboard website. Students are expected to watch the recorded lectures and engage with the assigned materials to grasp the course content.

Furthermore, students are required to complete designated course tasks. The instructor as well as the Learning Assistants will hold student/tutoring hours on Zoom to help with student questions on the learning material as well as course tasks. It is highly recommended that students take advantage of these student/tutoring hours to seek clarification on course content or assignments.

COMMUNICATION

- The primary method by which course-related information will be communicated is through Blackboard Course Announcements and emails.
- Reminders and notification of any schedule or assignment changes will be communicated through NDSU email and posted on the Blackboard announcements page.
- Please note that your NDSU email address is the official route for information.

REQUIRED STUDENT RESOURCES

Required Textbooks: None

We will not be using any textbooks in this course. We will rely on the official documentation that is available online for each of the frameworks and technologies that we will be using in this course. The instructor would provide the course material including PowerPoint slides, and web links to

Internet sources for the topics that will be covered. Students are strongly encouraged to take class notes.

Environment and Tools:

The languages, frameworks, and tools that will be introduced include the following:

- Web Application Frameworks:
 - Client Application Framework – Angular (TypeScript)
 - Server-Side Framework – Django (Python)
- Web Service – REST API
- Database – Google Firebase
- Single Page Application Design (SPA)
- Project Management – Trello
- Others – VS Code, JSON, Git, GitHub, HTML, DOM, Bootstrap, Angular Material, Stackblitz

Please note that each of the above frameworks and tools is open-source or available for free for student use.

EVALUATION PROCEDURES AND CRITERIA

The coursework would consist of assignments, exams, tutorials and a team project deliverable.

Assignments –

- There will be six mandatory Assignments to be completed and submitted during the course.
- Each assignment would be submitted electronically on the Blackboard for grading. Home assignments submitted via email will not be accepted.
- **Late work** on Assignments may be turned up to **4 days** after the due date and will incur a 20% penalty for each day late.
 - Due dates may be extended by the instructor.
 - Changes in due dates will be announced in class and posted as an announcement on Blackboard.
 - In the case of a university-sponsored activity, a required business trip, or medical procedure, a signed letter from your manager or doctor is required. In other circumstances, documentation will certainly help your case. Extensions on due dates will be at the discretion of the instructor.
- Students can discuss the assignment and exchange ideas but are required to complete the assignments on their own.

Exams –

- The course includes two exams, each assessing various aspects of understanding.
- **Each exam will consist of two parts.**
 - **Part 1** will include multiple-choice, short-answer, programming, and conceptual questions.
 - **Part 2** will require students to design and complete an Angular or Django project.

- **Exam 1** covers Weeks 1-8, held in Week 9 (refer to schedule).
- **Exam 2** covers Weeks 10-16, during the last week of the semester.
- **Please note that there is no Comprehensive Final Exam in the course.**
- Notify the instructor in advance if you can't attend an exam. Makeup exams, potentially essay and code writing, are at the instructor's discretion. For university-sponsored activities, business trips, or medical procedures, a signed letter is required. Documentation strengthens other cases.

Tutorials –

- There will be number of learning activities/tutorials that students are expected to submit periodically. The purpose of these activities is to demonstrate that students are learning the material that is provided in the course.

Team Project –

- Students will work in teams to build a Web application using the frameworks, tools, languages, and other technologies covered in the class.
- During week 1- 2 of the semester, students will be expected to self-organize themselves into teams of up to 4 students each. Team size cannot be larger than 4 students unless permitted by the instructor. Please contact the instructor at the earliest if you are unable to join any team. The instructor will help you join a team.
- Each team would be expected to propose a web application project that they would like to work on. The scope of the project should be sufficient to demonstrate the software engineering concepts and principles that are covered in the class and implementable by the end of the semester.
- Each team would get their Team Project Proposal approved by the instructor before they can proceed with the project. The instructor may suggest necessary changes to the project idea and scope before approving the project.
- Team Project Deliverables
 - Project Requirements –
 - First deliverable of the team project
 - Each team will document the project goals and requirements in the form of user stories and use cases.
 - Project Design –
 - Each team would document their design using wireframe mockups and other design techniques covered during the course.
 - Final Project Implementation, Demonstration and Presentation
 - Each team will implement and maintain the project in **GitHub**, demonstrate and present their project during the last week of the semester.
 - Each member of the team is required to be part of the presentation, provide details on their contribution to the project, and discuss the lessons learned from being on a team project.
- Note that each member of the team will also complete an anonymous peer review of other members of the team for each of the Team Project Deliverables.

- Grading for the team project would be based on the instructor's satisfaction in terms of the final project meeting the project's goals, requirements, design laid out in earlier deliverables, adherence to good agile development practices, processes, and peer reviews by team members.
- **Late submission of project deliverables** may be allowed up to **4 days** after the due date and will incur a 20% penalty for each day late.
 - Due dates may be extended by the instructor.
 - Changes in due dates will be announced in class and posted as an announcement on Blackboard.

Please Note:

If you dispute about the grading of any material, you have **two weeks** from the date the grade is *recorded* to request a change in the grade. After this time, no alterations will be considered, unless a grading error is discovered.

Coursework components and their weights on final grade:

- **Assignments** – 30% of Final Grade
 - *Assignment – 1: 5% of final grade*
 - *Assignment – 2: 5% of final grade*
 - *Assignment – 3: 5% of final grade*
 - *Assignment – 4: 5% of final grade*
 - *Assignment – 5: 5% of final grade*
 - *Assignment – 6: 5% of the final grade*
- **Tutorials** – 10% of Final Grade
 - There will be number of learning activities/tutorials that students are expected to submit periodically. These activities will be weighted equally.
- **Exams** – 30% of Final Grade
 - *Exam – 1: 15% of final grade*
 - *Exam – 2: 15% of final grade*
- **Team Project** – 30% of Final Grade
 - *Project Requirement: 5% of final grade*
 - *Project Design: 10% of final grade*
 - *Final Project and Presentation: 15 % of final grade*

The final letter grade would be based on the standard percentage grading system:

Greater than or equal to 90% = A

Less than 90 and greater than or equal to 80% = B,

Less than 80 and greater than or equal to 70% = C,

Less than 70 and greater than or equal to 60% = D,

Less than 60% = F.

ATTENDANCE STATEMENT

According to [NDSU Policy 333 \(www.ndsu.edu/fileadmin/policy/333.pdf\)](http://www.ndsu.edu/fileadmin/policy/333.pdf), attendance in classes is expected. **Please note that this is an online asynchronous class which means there are no meeting times so there is no expectation of any attendance. But students are required to complete their course tasks and submit them on time.** In the case of a university-sponsored activity, a required business trip or medical procedure, a signed letter from your manager or doctor is required. Please email me as far in advance of the expected absence to arrange any make-up or extensions. If you are unable to submit your work due to illness, email me within 24 hours of accommodation. Unexcused work cannot be made up.

MILITARY PERSONNEL AND VETERANS

Veterans and student service members with special circumstances or who are activated are encouraged to notify the instructor as soon as possible and are encouraged to provide Activation Orders.

AMERICANS WITH DISABILITIES ACT FOR STUDENTS WITH SPECIAL NEEDS STATEMENT

Any students with disabilities who need accommodations in this course are invited to share these concerns or requests with the instructor and contact the Center for Accessibility and Disability Resources as soon as possible.

FAMILY EDUCATIONAL RIGHTS AND PRIVACY ACT (FERPA) STATEMENT

Your personally identifiable information and educational records as they relate to this course are subject to [FERPA](#).

ACADEMIC HONESTY STATEMENT

The academic community is operated on the basis of honesty, integrity, and fair play. [NDSU Policy 335: Code of Academic Responsibility and Conduct](#) applies to cases in which cheating, plagiarism, or other academic misconduct have occurred in an instructional context. Students found guilty of academic misconduct are subject to penalties, up to and possibly including suspension and/or expulsion. Student academic misconduct records are maintained by the [Office of Provost](#). Informational resources about academic honesty for students and instructional staff members can be found at [Standards for Academic Honesty & Integrity at NDSU: Student Resources](#) and [Standards for Academic Honesty & Integrity at NDSU: Faculty](#).

COURSE SCHEDULE

This is a **tentative schedule** that is subject to change at the instructor's discretion. Any changes and specific dates for coursework submissions will be communicated in class as well as announcements through Blackboard and emails.

Week	Dates	Topic	Coursework Due
1	1/12 – 1/18	Agile and Scrum User Stories	
2	1/19 – 1/25	Web Concepts Git and GitHub	<i>Project Teams formed.</i>
3	1/26 – 2/1	HTML CSS Bootstrap DOM	Assignment – 1
4	2/2 – 2/8	TypeScript Angular	<i>Team Project Proposal</i>
5	2/9 – 2/15	Angular	Assignment - 2
6	2/16 – 2/22	Angular	
7	2/23 – 3/1	Angular	<i>Team Project Requirement</i>
8	3/2 – 3/8	Angular	Assignment - 3
SPRING BREAK (3/9 – 3/15)			
9	3/16 – 3/22	Angular	Exam - 1
10	3/23 – 3/29	Angular	Assignment – 4
11	3/30 – 4/5	Python Django	<i>Team Project Design</i>
12	4/6 – 4/12	Django	Assignment - 5
13	4/13 – 4/19	Django	
14	4/20 – 4/26	Django	
15	4/27 – 5/3	Software Architecture	Assignment - 6
16	5/4 – 5/10	Team Presentations	<i>Final Team Projects</i>
		Exam - 2	<i>TBD</i>

